Chapter 8. Conditional Independence and Gaussian Graphical Models

Graphical models for discovering relations, Lecture 1 (partial correlation, precision matrix, GGM, graphical lasso, neighbourhood selection)

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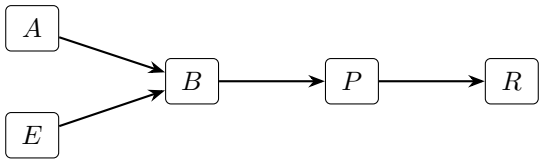
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Roadmap of the causal-discovery part

Lecture	Condition on the data	Relation to be discovered
1 (Ch. 8)	Gaussian observational data	undirected graph of conditional relations
2 (Ch. 9)	linear SEM with given directions	conditional independences a graph implies; limits of direction
3 (Ch. 10)	all main variables observed	the equivalence class of DAGs that data can distinguish
4 (Ch. 11)	hidden common causes allowed	latent confounding, relations and directions compatible with it
5 (Ch. 12)	extra assumption on the error distribution	more specific directions inside the equivalence class
6 (Ch. 13)	further identifiable models	directions for which causal discovery is possible

A running toy health example

The same five-variable health data are reused through all six lectures. The data-generating structure is



A	age	E	amount of exercise
B	body-mass index (BMI)	P	blood pressure
R	disease risk		

i.e. $E \to B \to P \to R$ and $A \to B$. Lecture 1 does not use directions and looks for conditional relations given the remaining variables; Lectures 2–3 study how directions are represented and how far they can be discovered; Lecture 4 adds a hidden lifestyle variable U, and Lecture 6 adds measurement error.

1 Correlation Mixes Direct and Indirect Relations

Let three standardised Gaussian variables be generated by

$$X_1 \sim N(0,1), \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.8X_2 + \epsilon_3, \qquad \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0, 0.36),$$

with $X_1, \epsilon_2, \epsilon_3$ mutually independent, so all three variances equal $0.64 + 0.36 = 1$.

Pair	Correlation	Why the relation arises
X_1, X_2	0.8	X_1 and X_2 are directly connected
X_2, X_3	0.8	X_2 and X_3 are directly connected
X_1, X_3	0.64	the relation is transmitted through X_2

Indeed $\mathrm{Cov}(X_1, X_3) = \mathrm{Cov}(X_1, 0.64X_1 + 0.8\epsilon_2 + \epsilon_3) = 0.64$. The right question is not "is there a correlation?" but "**does a relation remain after accounting for the other variables?**"

2 Conditional Independence

Definition: conditional independence

$X \perp\!\!\!\perp Y \mid Z$ if $p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$ for all z, equivalently $p(x \mid y, z) = p(x \mid z)$: among cases with the same Z, knowing Y gives no further information about X.

Marginal relation: $X_1 \not\perp\!\!\!\perp X_3$ — knowing X_1 changes what we know about X_3 (here Corr $= 0.64$). Conditional relation: $X_1 \perp\!\!\!\perp X_3 \mid X_2$ — comparing only cases with the same X_2, X_1 gives no additional information about X_3:

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Conditional independence does not fix causal direction, but it **narrows the candidate edges** of conditional dependence among variables.

3 Partial Correlation

3.1 Definition by residuals

Adjusting X_1 and X_3 for X_2:

$$X_1 = \alpha_1 + \beta_1 X_2 + r_1, \qquad X_3 = \alpha_3 + \beta_3 X_2 + r_3.$$

Quantity	Meaning
r_1	the part of X_1 not explained by X_2
r_3	the part of X_3 not explained by X_2
$\mathrm{Corr}(r_1, r_3)$	the partial correlation of X_1, X_3 given X_2

$$\rho_{13\cdot 2} = \mathrm{Corr}(r_1, r_3),$$

with population (linear-projection) residuals. For a set S, $\rho_{jk\cdot S}$ is defined in the same way after regressing both variables on X_S.

Theorem: first-order partial correlation

$$\rho_{13\cdot 2} = \frac{\rho_{13} - \rho_{12}\rho_{23}}{\sqrt{(1-\rho_{12}^2)(1-\rho_{23}^2)}}.$$

Proof. Standardise all variables. The projection coefficients are $\beta_1 = \rho_{12}$, $\beta_3 = \rho_{23}$, so $r_1 = X_1 - \rho_{12}X_2$ and $r_3 = X_3 - \rho_{23}X_2$. Then $\mathrm{Cov}(r_1, r_3) = \rho_{13} - \rho_{12}\rho_{23} - \rho_{23}\rho_{12} + \rho_{12}\rho_{23} = \rho_{13} - \rho_{12}\rho_{23}$, $\mathrm{Var}(r_1) = 1 - \rho_{12}^2$ and $\mathrm{Var}(r_3) = 1 - \rho_{23}^2$. ■

Example: a correlation of 0.64 can disappear

With $\rho_{12} = 0.8$, $\rho_{23} = 0.8$, $\rho_{13} = 0.64$,

$$\rho_{13\cdot 2} = \frac{0.64 - (0.8)(0.8)}{\sqrt{(1-0.8^2)(1-0.8^2)}} = 0.$$

Under joint Gaussianity we therefore conclude $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

4 The Precision Matrix

4.1 Zeros of the precision matrix are conditional independences

Let $X = (X_1, \dots, X_p)^\top \sim N_p(\mu, \Sigma)$ and call $\Omega = \Sigma^{-1}$ the **precision matrix**.

Precision matrix	Interpretation
$\Omega_{jk} = 0$	$X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$
$\Omega_{jk} \neq 0$	a relation between X_j, X_k remains given all the other variables

Theorem: precision matrix and full-order partial correlation

For $X \sim N_p(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Omega = \Sigma^{-1}$,

$$\rho_{jk\cdot -\{j,k\}} = -\frac{\Omega_{jk}}{\sqrt{\Omega_{jj}\Omega_{kk}}}, \qquad \text{so} \qquad \Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}.$$

Proof. Let $A = \{j, k\}$ and B the rest. For a Gaussian vector, $X_A \mid X_B$ is Gaussian with covariance equal to the Schur complement $\Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$. By the block-inversion formula this Schur complement equals $(\Omega_{AA})^{-1}$, where Ω_{AA} is the 2×2 block of Ω. Hence

$$\mathrm{Cov}(X_A \mid X_B) = \begin{pmatrix} \Omega_{jj} & \Omega_{jk} \\ \Omega_{jk} & \Omega_{kk} \end{pmatrix}^{-1} = \frac{1}{\Omega_{jj}\Omega_{kk} - \Omega_{jk}^2} \begin{pmatrix} \Omega_{kk} & -\Omega_{jk} \\ -\Omega_{jk} & \Omega_{jj} \end{pmatrix},$$

whose correlation is $-\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$. The conditional covariance does not depend on x_B, so this is the partial correlation. For a bivariate Gaussian, zero correlation is equivalent to independence. ■

One matrix inversion gives, for every pair simultaneously, the relation after accounting for all other variables.

4.2 Turning the precision matrix into a graph: the GGM

Definition: Gaussian graphical model

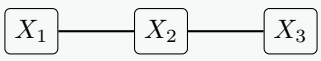
The GGM of $N_p(\mu, \Sigma)$ is the undirected graph on $\{1, \dots, p\}$ with

$$(j, k) \text{ is an edge} \iff \Omega_{jk} \neq 0.$$

Example: the three-variable chain

$$R = \begin{pmatrix} 1 & 0.8 & 0.64 \\ 0.8 & 1 & 0.8 \\ 0.64 & 0.8 & 1 \end{pmatrix}, \qquad \Omega = R^{-1} \approx \begin{pmatrix} 2.78 & -2.22 & 0 \\ -2.22 & 4.56 & -2.22 \\ 0 & -2.22 & 2.78 \end{pmatrix}.$$

Exactly: $\Omega_{11} = \Omega_{33} = 1/0.36$, $\Omega_{22} = 1.64/0.36$, $\Omega_{12} = \Omega_{23} = -0.8/0.36$, $\Omega_{13} = 0$ (check $R\Omega = I$). So the GGM is



The conditional relations $X_1 - X_2$ and $X_2 - X_3$ remain; $\Omega_{13} = 0$ removes the edge $X_1 - X_3$. The partial correlation of X_1, X_2 given X_3 is $0.8/\sqrt{1.64} \approx 0.62$, smaller than the marginal 0.8 (the same value follows from the first-order formula).

Remark: from pairwise to global statements

The definition uses the *pairwise* Markov property. For a positive density (e.g. a nondegenerate Gaussian) it is equivalent to the *global* Markov property: if a set S separates node sets A and B in the graph (every path from A to B passes through S), then $X_A \perp\!\!\!\perp X_B \mid X_S$ (Hammersley–Clifford). In the chain, $\{X_2\}$ separates X_1 from X_3.

5 Estimating the Graph

5.1 Inverting the sample covariance produces spurious edges

With the sample covariance S, the simplest estimate is $\widehat{\Omega} = S^{-1}$.

Data condition	Problem
finite sample	because of sampling error, entries that are truly 0 are estimated as small nonzero values
many variables	S^{-1} is unstable and most entries are nonzero
$p \geq n$	S is singular and the inverse may not exist

To discover relations we need a **sparse** precision-matrix estimate.

Method	Core idea	Suitable when
partial-correlation tests	compute partial correlations from S^{-1} and test $H_0 : \rho_{jk\cdot -\{j,k\}} = 0$	$p < n$ and sample large enough
graphical lasso	ℓ_1 penalty on the whole precision matrix, some entries estimated as 0	a sparse positive-definite precision matrix is needed
neighbourhood selection	Lasso regression of each variable on the rest; choose neighbours variable by variable	many variables, node-wise relations
CLIME	directly estimate a sparse Ω satisfying $S\Omega \approx I$	estimating the precision matrix in high dimension

The edge definition of the GGM is the same; the estimation method depends on the data conditions.

5.2 Testing a zero partial correlation

For Gaussian data, $H_0 : X_j \perp\!\!\!\perp X_k \mid X_S \iff H_0 : \rho_{jk\cdot S} = 0$. Let $\hat{\rho} = \hat{\rho}_{jk\cdot S}$ be the sample partial correlation and $q = |S|$.

Test	Statistic	Reference distribution
partial-correlation t-test	$T = \hat{\rho}\sqrt{\dfrac{n-q-2}{1-\hat{\rho}^2}}$	t_{n-q-2}
Fisher-z test	$Z = \sqrt{n-q-3}\,\operatorname{arctanh}(\hat{\rho})$	approximately $N(0,1)$
conditional distance correlation	$\mathrm{dCor}(X_j, X_k \mid X_{-\{j,k\}})$	p-value by resampling

Derivation: where the degrees of freedom come from

Regressing on q variables plus an intercept uses $q+1$ degrees of freedom, so the residual correlation behaves like an ordinary sample correlation from $n-q$ observations. For an ordinary correlation from m pairs, $T = r\sqrt{(m-2)/(1-r^2)} \sim t_{m-2}$ under $\rho = 0$,

and $\operatorname{arctanh}(r) \approx N(\operatorname{arctanh}\rho, 1/(m-3))$ by the delta method with variance stabilisation ($\operatorname{Var}(r) \approx (1-\rho^2)^2/m$ and $\frac{d}{d\rho}\operatorname{arctanh}\rho = 1/(1-\rho^2)$). Substituting $m = n - q$ gives the table.

Failing to reject H_0 does *not* prove conditional independence.

5.3 Graphical lasso

Derivation: Gaussian log-likelihood of Ω

For centred data, $\log p(x_i) = \frac{1}{2}\log\det\Omega - \frac{1}{2}x_i^\top\Omega x_i + c$, and $\sum_i x_i^\top\Omega x_i = \operatorname{tr}(\Omega\sum_i x_i x_i^\top) = n\operatorname{tr}(S\Omega)$. So the log-likelihood is $\frac{n}{2}\{\log\det\Omega - \operatorname{tr}(S\Omega)\} + c$, maximised at $\Omega = S^{-1}$ when $S \succ 0$.

The graphical lasso combines Gaussian fit with sparsity:

$$\widehat{\Omega}_\lambda = \arg\min_{\Omega\succ 0}\Big[\operatorname{tr}(S\Omega) - \log\det(\Omega) + \lambda\sum_{j\neq k}|\Omega_{jk}|\Big].$$

Part of the objective	Role
$\operatorname{tr}(S\Omega) - \log\det(\Omega)$	matches the covariance structure of the observed data
$\lambda\sum_{j\neq k}\lvert\Omega_{jk}\rvert$	sets small off-diagonal entries exactly to 0

The objective is convex in Ω; its optimality condition is $\Omega^{-1} - S - \lambda\Gamma = 0$ with $\Gamma_{jk} \in \partial|\Omega_{jk}|$ (and $\Gamma_{jj} = 0$). Consequently $|(\widehat{\Omega}^{-1})_{jk} - S_{jk}| \le \lambda$ for all $j \neq k$, with equality on the estimated edges. The solution is positive definite even when $p \ge n$. λ is chosen by CV log-likelihood, EBIC or stability selection.

5.4 Neighbourhood selection

Theorem: conditional distribution of one node

For centred $X \sim N_p(0, \Sigma)$ and $\Omega = \Sigma^{-1}$,

$$X_j \mid X_{-j} \sim N\Big(-\frac{\Omega_{j,-j}}{\Omega_{jj}}X_{-j},\ \frac{1}{\Omega_{jj}}\Big),$$

so the population coefficients of the regression of X_j on the other variables are $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, and $\beta_k^{(j)} = 0 \iff \Omega_{jk} = 0$.

Proof. Apply the Schur-complement argument with $A = \{j\}$: $\operatorname{Var}(X_j \mid X_{-j}) = (\Omega_{jj})^{-1}$, and the conditional mean $\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}X_{-j}$ equals $-\Omega_{jj}^{-1}\Omega_{j,-j}X_{-j}$ by the block-inverse identity $\Omega_{j,-j} = -\Omega_{jj}\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}$. ■

In practice the population coefficients are estimated sparsely by the Lasso:

$$\hat{\beta}^{(j)} = \arg\min_\beta\Big\{\frac{1}{2n}\|X_j - X_{-j}\beta\|_2^2 + \lambda\|\beta\|_1\Big\}.$$

If $\hat{\beta}_k^{(j)} \neq 0$, $j - k$ is an edge candidate. In samples the two regressions (j on k and k on j) may disagree, so they are combined by the **AND** rule (both nonzero) or the **OR** rule (at least one nonzero). Each regression costs one Lasso, which parallelises over nodes and scales to large p; the price is that the result is a graph, not a positive-definite $\widehat{\Omega}$.

5.5 CLIME

$$\widehat{\Omega} = \arg\min_{\Omega} \|\Omega\|_1 \quad \text{subject to} \quad \|S\Omega - I\|_\infty \leq \lambda,$$

a linear program that splits into one column at a time; the estimate is symmetrised afterwards. It estimates the sparsest matrix that is approximately an inverse of S.

6 GGM Edges Are Conditional Associations, Not Causal Directions

Interpretable from a GGM	Not interpretable from a GGM alone
Does a relation remain after accounting for the other observed variables?	Which variable is the cause and which the effect?
What are the sign and size of the partial correlation?	Is an edge a direct causal effect?
Which variables are conditionally connected to many others?	Is there a hidden common cause?

For example, $X_j - X_k$ means that the two variables remain related after accounting for the remaining observed variables. It means neither $X_j \to X_k$ nor $X_j \leftarrow X_k$. Worse, conditioning on a common *effect* can create an edge between two independent causes (Ch. 9). To infer directions we need DAGs and additional structure-learning assumptions.

Summary

- Correlation mixes direct and indirect relations; conditional independence asks what remains given the other variables.
- Partial correlations are correlations of residuals; for Gaussians, $\Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ and $\rho_{jk\cdot\text{rest}} = -\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$.
- The GGM puts an edge where $\Omega_{jk} \neq 0$; it is estimated by partial-correlation tests, graphical lasso, neighbourhood selection or CLIME.
- GGM edges are undirected conditional associations; directions require the DAG machinery of the next chapters.

Chapter 9. Gaussian Linear SEMs and Directed Relations

Graphical models, Lecture 2 (DAGs, chain/fork/collider, d-separation, faithfulness, Markov equivalence, CPDAG)

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1 From Undirected to Directed Graphs

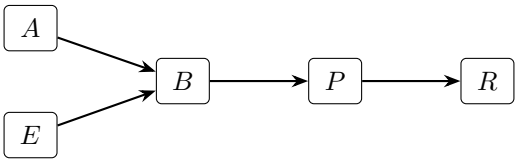
Lecture 1: conditional-dependence graph	Lecture 2: directed graph
$X_j - X_k \iff X_j \not\perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ finds relations that remain after accounting for the others edges but no directions	$X_k \to X_j$ expresses each arrow by a generating equation per variable reads the conditional independences the graph creates; checks which directions data cannot distinguish

The key question: **how far can the direction of relations be inferred from Gaussian observational data?**

2 A Gaussian Linear SEM Is a Bundle of Regressions on a DAG

2.1 The health example as equations

Consider the generating relations $A \to B \leftarrow E$ and $B \to P \to R$:



$$A = \epsilon_A, \quad E = \epsilon_E, \quad B = \beta_{AB}A + \beta_{EB}E + \epsilon_B, \quad P = \beta_{BP}B + \epsilon_P, \quad R = \beta_{PR}P + \epsilon_R,$$
$$\epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma_j^2).$$

Each variable is a linear combination of its parents plus its own error; with independent Gaussian errors the whole vector is multivariate Gaussian.

Definition: DAG and Gaussian linear SEM

A directed acyclic graph (DAG) G has directed edges and no directed cycle. With $\text{pa}(j)$ the parents of node j,

$$X_j = \sum_{k \in \text{pa}(j)} \beta_{kj} X_k + \epsilon_j, \qquad \epsilon_1, \ldots, \epsilon_p \text{ independent}, \ \epsilon_j \sim N(0, \sigma_j^2).$$

Variable	Parents	Local regression
A, E	none	$A = \epsilon_A$, $E = \epsilon_E$
B	A, E	$B = \beta_{AB}A + \beta_{EB}E + \epsilon_B$
P	B	$P = \beta_{BP}B + \epsilon_P$
R	P	$R = \beta_{PR}P + \epsilon_R$

For instance exercise E does not enter the equation of blood pressure P directly, but is related to P through the path $E \to B \to P$.

2.2 Matrix form and the implied covariance

Let B be the $p \times p$ matrix with $B_{jk} = \beta_{kj}$ (row j holds the coefficients of the equation of X_j) and $D = \mathrm{diag}(\sigma_1^2, \dots, \sigma_p^2)$. Then $X = BX + \epsilon$, and because G is acyclic, B can be permuted to be strictly lower triangular, so $I - B$ is invertible:

$$X = (I - B)^{-1}\epsilon, \qquad \Sigma = (I - B)^{-1} D (I - B)^{-\top}, \qquad \Omega = \Sigma^{-1} = (I - B)^{\top} D^{-1} (I - B).$$

The density factorises along the DAG (**Markov factorisation**):

$$p(x_1, \dots, x_p) = \prod_{j=1}^{p} p(x_j \mid x_{\mathrm{pa}(j)}).$$

Theorem: the GGM of an SEM is its moral graph

For $j \neq k$,

$$\Omega_{jk} = -\frac{B_{jk}}{\sigma_j^2} - \frac{B_{kj}}{\sigma_k^2} + \sum_{l \neq j,k} \frac{B_{lj} B_{lk}}{\sigma_l^2}.$$

Hence $\Omega_{jk} \neq 0$ only if j, k are adjacent in G or have a common child; generically (no cancellation) exactly then. The GGM of Ch. 8 is the **moral graph**: the skeleton plus edges "marrying" the parents of each child.

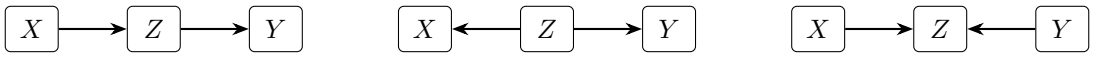
Proof. $\Omega_{jk} = \sum_l (I - B)_{lj} (I - B)_{lk} / \sigma_l^2$. The term $l = j$ gives $(1)(-B_{jk})/\sigma_j^2$, the term $l = k$ gives $(-B_{kj})(1)/\sigma_k^2$, and $l \neq j, k$ gives $B_{lj} B_{lk} / \sigma_l^2$, nonzero only if l is a child of both. ■

In the health example A and E are independent, yet the GGM contains $A - E$, because both are parents of B: conditioning on B (among all other variables) connects them. This is the first warning that undirected conditional relations are not causal edges.

3 Chains, Forks and Colliders

Structure	Graph	Marginally	Conditioning on Z
chain	$X \to Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
fork	$X \leftarrow Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
collider	$X \to Z \leftarrow Y$	$X \perp\!\!\!\perp Y$	X and Y related

These are the typical patterns observed when the Markov condition and faithfulness hold (§4). Conditioning **closes** chains and forks and **opens** colliders.



chain: open
given Z: blocked

fork: open
given Z: blocked

collider: blocked
given Z: open

Figure 1: The three elementary paths and what conditioning on the middle node does to the flow of association between X and Y. Only the collider reverses the pattern, which is why v-structures are detectable from data. Conditioning on a *descendant* of a collider also opens it.

Theorem: the three patterns in a Gaussian SEM

Let all errors be independent with variances 1 and coefficients $a, b \neq 0$.

1. Chain $Z = aX + \epsilon_Z$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.

2. Fork $X = aZ + \epsilon_X$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
3. Collider $Z = aX + bY + \epsilon_Z$ with $X \perp\!\!\!\perp Y$: $\mathrm{Cov}(X, Y) = 0$ but $\rho_{XY \cdot Z} = -\dfrac{ab}{\sqrt{(1+b^2)(1+a^2)}} \neq 0$.

Proof. (1) $Y = abX + b\epsilon_Z + \epsilon_Y$ so $\mathrm{Cov}(X, Y) = ab$; the residual of Y on Z is ϵ_Y, independent of (X, Z), so the partial correlation is 0. (2) $\mathrm{Cov}(X, Y) = ab\mathrm{Var}Z$; residuals of X, Y on Z are ϵ_X, ϵ_Y, independent. (3) $\mathrm{Cov}(X, Y) = 0$. With $\mathrm{Var}Z = a^2 + b^2 + 1$, $\mathrm{Cov}(X, Z) = a$, $\mathrm{Cov}(Y, Z) = b$, the residual covariance is $0 - ab/(a^2 + b^2 + 1)$ and the residual variances are $1 - a^2/(a^2 + b^2 + 1) = (1 + b^2)/(a^2 + b^2 + 1)$ and $(1 + a^2)/(a^2 + b^2 + 1)$; divide. ■

In a collider, telling us that the effect Z has a given value makes information about one cause change what we know about the other ("explaining away"): if BMI is high and age is low, exercise is probably low. **The change of conditional independence is the key information that distinguishes a collider from the other two structures.**

4 d-Separation, the Markov Condition and Faithfulness

Definition: d-separation

A path between X and Y is **blocked** by a set S if it contains (i) a non-collider (chain or fork node) that is in S, or (ii) a collider that is not in S and has no descendant in S. X and Y are d-separated by S, written $X \perp_G Y \mid S$, if every path between them is blocked.

Definition: causal Markov condition and faithfulness

Markov: $X \perp_G Y \mid S \Rightarrow X \perp\!\!\!\perp Y \mid S$ in the distribution (equivalent to the Markov factorisation). **Faithfulness**: the converse, $X \perp\!\!\!\perp Y \mid S \Rightarrow X \perp_G Y \mid S$: every conditional independence in the distribution is implied by the graph.

Under both, the conditional independences of the data are *exactly* the d-separations of G, which is what structure learning (Ch. 10) exploits.

Example: how faithfulness fails — cancellation

Let $X \sim N(0, 1)$, $Z = aX + \epsilon_Z$, $Y = cX + bZ + \epsilon_Y$. Then $\mathrm{Cov}(X, Y) = c + ab$. If $c = -ab$, the direct and indirect effects cancel and $X \perp\!\!\!\perp Y$ although $X \to Y$ is an edge. The set of such coefficient values has Lebesgue measure zero, which is the usual justification of faithfulness; near-cancellation, however, makes tests unreliable in finite samples.

Example: d-separation in the health example

$A \perp E$ (the only path $A \to B \leftarrow E$ has an unconditioned collider), but $A \not\perp E \mid P$ (P is a descendant of the collider B). $E \perp R \mid B$ and $E \perp R \mid P$ (the chain is blocked). $A \perp P \mid B$.

5 Markov Equivalence

5.1 Directions with the same conditional independences cannot be distinguished

The three DAGs

$$X_1 \to X_2 \to X_3, \qquad X_1 \leftarrow X_2 \to X_3, \qquad X_1 \leftarrow X_2 \leftarrow X_3$$

all imply the same conditional independence $X_1 \perp\!\!\!\perp X_3 \mid X_2$ and nothing else. Hence from observational data alone the three directions cannot be told apart. DAGs that imply the same conditional independences form a **Markov equivalence class**.

Theorem: the three DAGs define the same Gaussian model

A Gaussian distribution on (X_1, X_2, X_3) factorises according to one of the three DAGs iff $X_1 \perp\!\!\!\perp X_3 \mid X_2$; so all three describe exactly the same set of covariance matrices.

Proof. If $X_1 \perp\!\!\!\perp X_3 \mid X_2$ then $p = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_2)$ (first DAG). By Bayes' rule $p(x_1)p(x_2 \mid x_1) = p(x_2)p(x_1 \mid x_2)$, giving $p(x_2)p(x_1 \mid x_2)p(x_3 \mid x_2)$ (fork), and similarly $p(x_2 \mid x_3)p(x_3)$ replaces $p(x_2)p(x_3 \mid x_2)$ (third DAG). Conversely each factorisation implies the conditional independence. Each conditional of a Gaussian is a Gaussian linear regression, so every factorisation is a Gaussian linear SEM. ■

The collider $X_1 \to X_2 \leftarrow X_3$ has a different pattern, $X_1 \perp\!\!\!\perp X_3$ but $X_1 \not\perp\!\!\!\perp X_3 \mid X_2$, so it *can* be distinguished.

5.2 Skeleton and unshielded colliders

Definition: skeleton and v-structure

The **skeleton** is the adjacency structure left after deleting arrowheads: $X \to Z \to Y \mapsto X - Z - Y$. An **unshielded collider** (v-structure) is $X \to Z \leftarrow Y$ with X and Y non-adjacent.

Theorem: Verma–Pearl characterisation

Two DAGs are Markov equivalent iff they have the same skeleton and the same unshielded colliders.

Idea. Adjacency is detectable: X, Y are adjacent iff no set d-separates them (an edge is never blocked; for non-adjacent nodes, the parents of the later one in a topological order d-separate them). Given the skeleton, an unshielded triple $X - Z - Y$ is a collider iff Z is *not* in any set separating X and Y, which again is a CI statement. Conversely, if skeletons and v-structures agree, one shows that every path is blocked by the same sets in both graphs. ■

Without further distributional assumptions, observational data usually identify not one DAG but this equivalence class.

5.3 CPDAG

Definition: CPDAG

The completed partially directed acyclic graph of an equivalence class has a directed edge $X \to Y$ if every DAG in the class contains $X \to Y$, and an undirected edge $X - Y$ if the direction varies within the class.

For instance the CPDAG $X - Z - Y$ represents $X \to Z \to Y$, $X \leftarrow Z \to Y$ and $X \leftarrow Z \leftarrow Y$ together; the collider $X \to Z \leftarrow Y$ is its own class and its CPDAG has both arrows. For the health example, the CPDAG is $A \to B \leftarrow E$, $B \to P \to R$: the v-structure at B fixes $A \to B$ and $E \to B$, and then $B \to P$ is forced (otherwise $A \to B \leftarrow P$ would be a new v-structure), and likewise $P \to R$. **An undirected edge does not mean "no relation"; it means the direction could not be distinguished.**

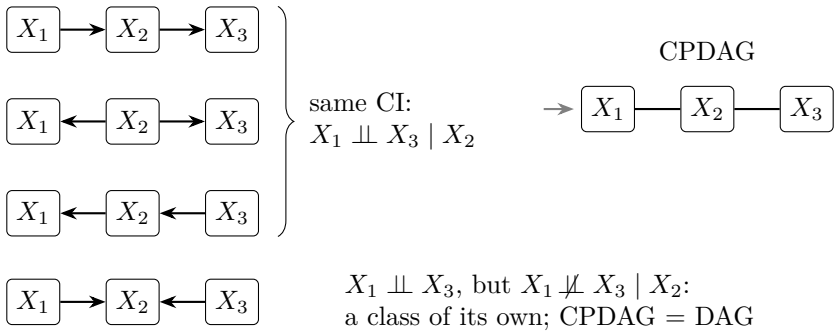


Figure 2: Markov equivalence. The chain, fork and reversed chain share the skeleton and have no v-structure, so they imply the same conditional independences; their CPDAG keeps both edges undirected. The collider has a v-structure and is identified.

Summary

- A Gaussian linear SEM writes each variable as a regression on its parents in a DAG; $\Sigma = (I - B)^{-1} D (I - B)^{-\top}$ and the GGM is the moral graph.
- Chains and forks are closed by conditioning on the middle node; colliders are opened. d-separation formalises this.
- Under the Markov condition and faithfulness, conditional independences equal d-separations.
- Markov-equivalent DAGs share skeleton and v-structures; Gaussian observational data identify only this class, summarised by the CPDAG.

The key question of the next chapters: in Gaussian observational data, **how far can the direction of relations be inferred**, and with which algorithms?

Chapter 10. Structure Learning: PC, GES and MMHC

Graphical models, Lecture 3 (constraint-based and score-based discovery of the Markov equivalence class)

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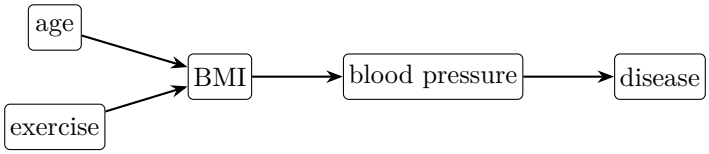
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6 Comparison **6**

1 What Observational Data Can Reveal

Questions of this lecture: which relations remain after accounting for the other variables? Which edges does conditional independence remove? How far can directions be fixed from observational data alone?

The toy example used throughout:



In a real analysis this graph is unknown and relations are searched for using observational data only. **The output of structure learning is the set of relations that the assumptions and the data allow; observational data do not fix every arrow.**

2 Assumptions Matter as Much as Data

Assumption	Practical meaning
acyclicity	relations at one time point are represented by a DAG without cycles
causal Markov condition	paths blocked in the graph appear as conditional independences in the data
faithfulness	no accidental cancellation of coefficients makes relations look independent
causal sufficiency	no unobserved common cause of two analysed variables
same generating process	all samples come from the same system, without selection bias

PC and GES rely in particular on causal sufficiency. If hidden common causes are suspected, FCI is needed (Ch. 11).

3 The PC Algorithm

3.1 Skeleton phase

Algorithm: PC, skeleton

1. Connect all pairs of variables by undirected edges.
2. For $\ell = 0, 1, 2, \ldots$: for each adjacent pair (X_j, X_k) and each conditioning set $S \subseteq \text{adj}(X_j) \setminus \{X_k\}$ of size ℓ, test
$$H_0 : X_j \perp\!\!\!\perp X_k \mid X_S.$$
3. If H_0 is not rejected for some S, remove the edge $X_j - X_k$,
4. and store that S as the separating set $\text{Sep}(X_j, X_k)$.

Stop when no node has more than ℓ remaining neighbours.

The tests must suit the data: Gaussian continuous data use partial correlation with the Fisher-z test (Ch. 8); categorical data use the χ^2 or G^2 test; nonlinear relations need nonlinear CI tests (Ch. 11).

The number of remaining edges depends on the significance level and sample size. The first output of PC is an undirected **skeleton**; an edge means a relation that remains after accounting for the other variables.

Example: skeleton of the toy example

Pair	Conditioning set	Decision	Action
age, exercise	$\varnothing$	independent	remove edge
age, blood pressure	BMI	conditionally independent	remove edge
exercise, blood pressure	BMI	conditionally independent	remove edge
BMI, disease	blood pressure	conditionally independent	remove edge

(Pairs such as age–disease and exercise–disease are removed similarly, e.g. given BMI.) The remaining skeleton is

$$\mathcal{E}_{\text{skel}} = \{\text{age–BMI},\ \text{exercise–BMI},\ \text{BMI–blood pressure},\ \text{blood pressure–disease}\}.$$

At this stage only the backbone of conditional relations is left; no direction has been decided yet.

3.2 Orientation phase

Colliders tell directions from observational data. The edge between age and exercise was removed with no conditioning set: $\text{Sep}(\text{age}, \text{exercise}) = \varnothing$. Both variables are adjacent to BMI, they are not adjacent to each other, and BMI is not in their separating set. So PC orients

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}.$$

Before conditioning on BMI, age and exercise are independent; given BMI they can become dependent. If two non-adjacent variables point into the same middle variable, the middle variable is a collider.

Algorithm: PC, orientation

1. **v-structures**: for every unshielded triple $X - Z - Y$ (X, Y non-adjacent) with $Z \notin \text{Sep}(X, Y)$, orient $X \to Z \leftarrow Y$.
2. **Meek rules**, applied until nothing changes:

   R1 (no new collider) $X \to Z - Y$, X, Y non-adjacent $\Rightarrow Z \to Y$;

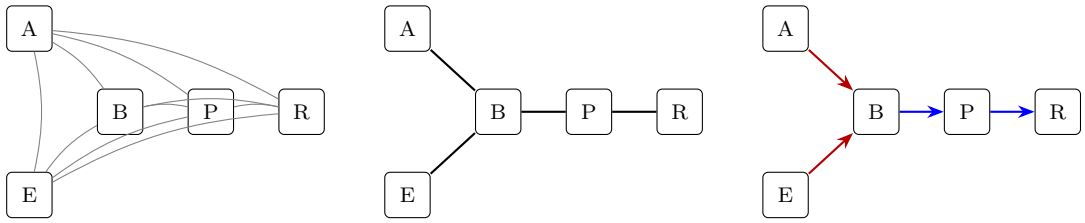
   R2 (no cycle) $X \to Z \to Y$ and $X - Y \Rightarrow X \to Y$;

   R3 $X - Z_1 \to Y$, $X - Z_2 \to Y$, $X - Y$, Z_1, Z_2 non-adjacent $\Rightarrow X \to Y$.

Preventing new colliders: if $A \to B - C$ and $A \not\sim C$, orienting $C \to B$ would create $A \to B \leftarrow C$, which the data did not support; hence $B \to C$. **Preventing cycles**: if $A \to B \to C$ and $A - C$, then $C \to A$ would create a directed cycle; hence $A \to C$. In the example the following directions are fixed in turn:

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}, \qquad \text{BMI} \to \text{blood pressure} \to \text{disease}.$$

PC directions are interpreted only within the range forced by the conditional-independence pattern and the assumptions.



(1) complete graph (2) skeleton after CI tests (3) v-structure + Meek R1

Figure 1: PC on the toy example (A age, E exercise, B BMI, P blood pressure, R disease). Step 2 removes every edge whose endpoints become conditionally independent. Step 3 orients the v-structure $A \to B \leftarrow E$ (red, since $B \notin \mathrm{Sep}(A, E) = \varnothing$) and then propagates $B \to P \to R$ (blue) to avoid creating new colliders.

3.3 Why PC is correct

Theorem: correctness of PC with a CI oracle

Assume the Markov condition, faithfulness and causal sufficiency, and that every CI query is answered correctly. Then PC returns the CPDAG of the true DAG.

Proof. *Adjacencies.* An edge of G is never removed: by faithfulness, adjacent variables are dependent given every set. If X, Y are non-adjacent, one of them, say Y, is not a descendant of the other; by the local Markov property $X \perp\!\!\!\perp Y \mid \mathrm{pa}(X)$. Since true edges are never removed, $\mathrm{pa}(X) \subseteq \mathrm{adj}(X)$ at every stage, so this set (or another separating set) is found when $\ell = |\mathrm{pa}(X)|$. *v-structures.* Take an unshielded triple $X - Z - Y$. If Z is a non-collider on $X \cdot Z \cdot Y$, every set separating X, Y must contain Z (else that path is open); if Z is a collider, no separating set contains Z (conditioning on Z opens the path). So the single stored set $\mathrm{Sep}(X, Y)$ decides correctly. *Remaining edges.* The Meek rules are sound (each alternative orientation creates a new v-structure or a cycle) and complete for this situation (Meek, 1995), giving the CPDAG. ■

Remark: finite-sample issues

With real tests, an early type-II error removes an edge that later cannot be recovered and changes the separating sets used for orientation; errors propagate. The output also depends on the order in which pairs are tested; *PC-stable* removes this order dependence by updating adjacencies only after each level ℓ. Conflicting v-structures (both $X \to Z \leftarrow Y$ and $Z \to Y$ demanded) signal test errors or violated assumptions.

4 Score-Based Search

4.1 BIC for Gaussian linear SEMs

Score-based search compares candidate graphs by goodness of fit and complexity. For a candidate DAG G,

$$S_{\mathrm{BIC}}(G) = \ell(\hat{\theta}_G) - \frac{d_G}{2} \log n,$$

where $\ell(\hat{\theta}_G)$ measures how well the graph explains the data and d_G is the number of estimated parameters. Adding an edge can improve fit but increases the complexity penalty.

Theorem: BIC decomposes into local scores

For a Gaussian linear SEM,

$$S_{\mathrm{BIC}}(G) = \sum_{j=1}^{p} S_j\{X_j, \mathrm{pa}_G(j)\}, \qquad S_j = -\frac{n}{2}\log \hat{\sigma}^2_{j|\mathrm{pa}(j)} - \frac{|\mathrm{pa}(j)|+1}{2}\log n + c,$$

where $\hat{\sigma}^2_{j|\mathrm{pa}(j)}$ is the residual variance of the least-squares regression of X_j on its parents.

Proof. By the Markov factorisation the log-likelihood is a sum of the log-likelihoods of the local regressions, whose parameters are separate. The maximised Gaussian log-likelihood of one regression is $-\frac{n}{2}\log(2\pi\hat{\sigma}^2) - \frac{n}{2}$, and that regression has $|\mathrm{pa}(j)|$ coefficients plus one variance (centred data). ■

Decomposability means that a local move (adding, deleting or reversing one edge) changes only one or two local scores, which makes search cheap. Markov-equivalent DAGs give the same maximum likelihood and the same number of parameters, so BIC is **score equivalent**: score-based methods also cannot distinguish directions within a Markov equivalence class from data alone.

4.2 GES and hill climbing

	GES	Hill climbing
search space	CPDAGs representing Markov equivalence classes	individual DAGs
moves	forward phase adds relations, backward phase removes them	add, delete or reverse an edge
typical output	a high-scoring CPDAG	one high-scoring DAG
caution	remaining undirected edges are undetermined	not every output direction is identified

Algorithm: Greedy Equivalence Search

Forward: start from the empty graph; repeatedly apply the single edge insertion (in equivalence-class space) that increases the score most, until no insertion helps. **Backward**: from there, repeatedly apply the best edge deletion until no deletion helps. Output the CPDAG.

Theorem: consistency of GES (Chickering, 2002)

If the distribution is faithful to a DAG and the score is consistent and decomposable (e.g. BIC), then GES returns the true CPDAG with probability tending to 1 as $n \to \infty$, despite being greedy.

Hill climbing over DAGs can get stuck in local optima; tabu lists and random restarts help. Computing a DAG as output and identifying arrows from data are two different things.

5 Computational Burden and Hybrid Methods

	Why the cost grows	Remaining problem
PC	repeats CI tests over many conditioning sets	tests multiply when the graph is dense; early test errors propagate to later edges and directions
GES	compares scores of many edge additions/deletions at each step	with many variables score computations grow; greedy results depend on sample and search path

For a maximal degree k, the number of PC tests is polynomial in p (roughly of order p^{k+2}), but it is exponential in k. Ways to reduce the burden:

- **MMHC**: reduce candidate edges by conditional-independence tests, then run a score search only within the restricted space;
- **FLOP**: search efficiently over variable orders and parent sets to cut the cost of score computations.

5.1 MMHC

CI tests $\longrightarrow$ candidate skeleton $\longrightarrow$ score search $\longrightarrow$ DAG candidate

Stage 1 (candidate relations): for each variable, find the neighbours it may be directly connected to by CI tests (Max-Min Parents and Children); keep only those pairs as candidate edges. **Stage 2 (restricted score search)**: within the candidate edges, repeat additions, deletions and reversals, and select a DAG with high BIC or a similar score. An edge dropped in the first stage cannot be restored in the second, and not every direction of the output DAG is identified.

6 Comparison

	PC	GES	MMHC
approach	constraint (CI) based	score based	both, in sequence
search	removes edges when independent	changes the CPDAG to increase the score	restricts candidates by tests, then score search over DAGs
typical output	CPDAG	CPDAG	DAG candidate
key settings	independence test, significance level	score, complexity penalty	tests, significance level, score
sensitive to	test errors	score and search path	edges lost in stage 1

Summary

- Under acyclicity, Markov, faithfulness and causal sufficiency, observational data identify the Markov equivalence class.
- PC removes edges by CI tests, orients v-structures from the separating sets, then propagates orientations with the Meek rules; it is correct with a CI oracle.
- BIC decomposes into local regression scores and is score equivalent; GES searches equivalence classes and is consistent.

- MMHC combines both; its first-stage errors are irreversible.

Chapter 11. Relational Inference Matched to the Data Structure

Graphical models, Lecture 4 (hidden confounders, FCI and PAGs, nonlinear CI tests, PCMCI+, feedback)

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1 PC Assumes All Common Causes Are Observed

Premises used by PC: relations among the observed variables are represented by one DAG; no common cause of the analysed variables is missing; the Markov condition and faithfulness hold and conditional independences are tested properly.

Problem: an unobserved common cause. Suppose $E \leftarrow U \rightarrow P$ with U a hidden lifestyle factor. The observed data contain only exercise E and blood pressure P. E and P are related, but they are not directly connected. Because PC assumes all common causes are observed, it may represent this relation as a direct edge. **If hidden common causes are possible, an edge found by PC cannot be interpreted as a direct relation.**

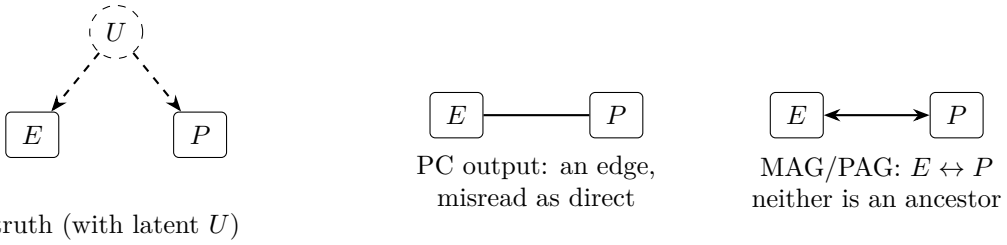


Figure 1: A hidden common cause. Marginalising U leaves a dependence between E and P with no direct effect. A DAG over the observed variables must pick a direction; the ancestral-graph language of FCI has a bidirected edge for exactly this situation (identifiable when enough other variables are observed).

1.1 Observational data may not distinguish three explanations

Suppose only X and Y are observed and they are related.

Possible structure	Explanation	Observed conditional independences
$X \rightarrow Y$	X causes Y	none
$X \leftarrow Y$	Y causes X	none
$X \leftarrow U \rightarrow Y$	hidden common cause U	none

Additional observed variables and their conditional independences can exclude some explanations, but a single DAG is not always determined.

2 FCI Infers a PAG Instead of a DAG

2.1 Ancestral graphs and PAGs

When latent variables are marginalised out of a DAG, the conditional independences among the observed variables are represented by a **maximal ancestral graph** (MAG) with directed ($\rightarrow$) and bidirected ($\leftrightarrow$) edges: $X \rightarrow Y$ means X is an ancestor of Y; $X \leftrightarrow Y$ means neither is an ancestor of the other (a latent confounder). Markov-equivalent MAGs are summarised by a **partial ancestral graph** (PAG), in which an endpoint mark is shown only if it is common to all MAGs of the class; otherwise a circle $\circ$ is drawn.

	Object represented	Meaning of relations
PC's CPDAG	DAG equivalence class under the assumption of no missing common causes among analysed variables	only directions common to all equivalent DAGs are fixed
FCI's PAG	equivalence class of observed-variable structures allowing hidden common causes	only endpoints common to all compatible structures are fixed

FCI (Fast Causal Inference) uses conditional independences but does *not* assume causal sufficiency. The circles and arrowheads of a PAG are not an incomplete result: they display the range that observational data can distinguish.

2.2 Reading PAG endpoints

PAG edge	How to read it
$X \to Y$	apart from selection bias, in every compatible structure X is an ancestor of Y and Y is not an ancestor of X
$X \leftrightarrow Y$	neither is an ancestor of the other; compatible with a structure with a hidden common cause
$X \circ\!\!\to Y$	it is certain that Y is not an ancestor of X; the mark at X is undetermined
$X \circ\!\!-\!\!\circ Y$	the two variables are adjacent but both endpoints are undetermined

An arrowhead means that variable is *not an ancestor* of the other. A circle may be a tail or an arrowhead depending on the compatible structure. $X \to Y$ in a PAG is an ancestral relation, not the single direct effect of a latent DAG.

2.3 FCI can still find the arrowheads of colliders

Suppose CI tests give the skeleton $X - Y - Z$ with $X \not\sim Z$, and $X \perp\!\!\!\perp Z$ unconditionally but $X \not\perp\!\!\!\perp Z \mid Y$. Without conditioning on Y, the path is blocked at the collider and X, Z are independent; conditioning on the collider opens the path and they become related. Hence FCI orients the arrowheads into Y:

$$X \circ\!\!\to Y \leftarrow\!\!\circ Z.$$

Whether X (or Z) is a cause of Y or shares a hidden cause with it remains open.

2.4 FCI checks wider separating sets, then orients conservatively

Algorithm: FCI

Edge removal.

1. Start from the complete graph on the observed variables.
2. If $X \perp\!\!\!\perp Y \mid S$ is found, remove $X - Y$ (as in PC's skeleton phase).
3. Widen the candidate separating sets beyond current neighbours to the variables on possible paths (Possible-D-Sep) and check again: because of hidden variables, a separating set may not lie among the current neighbours.

Endpoint determination.

1. Record the separating set used when an edge was removed.
2. Mark as colliders the middle variables not contained in the separating set.

3. Propagate only those endpoints that are forced, using the colliders and the PAG orientation rules (Zhang's rules R1–R10).

Undetermined endpoints remain circles $\circ$.

Remark: why neighbours are not enough

In a DAG, a non-adjacent pair is always separated by the parents of one of them, which are neighbours. In a MAG this fails: a separating set may have to contain a node W that reaches X only through a path whose inner nodes are all colliders (e.g. $W \leftrightarrow V \leftrightarrow X$ created by latent variables), and such a W need not be adjacent to X or Y. The set D-Sep(X, Y) collects such nodes; since it depends on the unknown MAG, FCI uses the computable superset Possible-D-Sep from the intermediate graph. PC, which only searches among neighbours, would wrongly keep the edge. The extra tests are the price of not assuming causal sufficiency.

Like PC, FCI starts from conditional independences, but checks the additional separation possibilities needed because of hidden variables. **RFCI** (really fast CI) skips most of the Possible-D-Sep search and runs targeted extra tests on triples, reducing computation at the cost of a slightly less informative (but still sound) output.

3 When Basic Assumptions Break, Methods Change

Basic assumption	Situation where it fails	Candidate methods
all common causes observed	unobserved common causes exist	FCI, RFCI
the chosen test captures the relation	partial correlation misses nonlinear relations	KCI, distance correlation
independent observations	autocorrelation and time lags	PCMCI+
no cycles in the graph	feedback among variables	CCD, cyclic LiNGAM

The choice of algorithm depends less on the data size than on **which basic assumptions can be maintained**.

3.1 Nonlinear relations: change the independence test

Partial correlation measures linear relations.

Example: dependence with zero correlation

Let $X \sim \text{Unif}(-1, 1)$ and $Y = X^2 + \varepsilon$ with $\varepsilon \perp\!\!\!\perp X$. Then $\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ because the distribution of X is symmetric, although Y depends strongly on X. A Fisher-z test would declare them independent and PC would remove the edge.

CI test	When to use
Fisher-z	linear conditional relations in continuous Gaussian data
kernel CI test (KCI)	general nonlinear conditional relations (based on cross-covariance operators in reproducing-kernel spaces)
distance correlation	nonlinear relations without strong assumptions on the distribution; $\text{dCor}(X, Y) = 0$ iff $X \perp\!\!\!\perp Y$

The search procedures of PC and FCI stay the same; only the CI test is replaced. Nonparametric tests are more flexible but lose power with small samples or many conditioning variables, and are computationally heavier.

3.2 Time series: PCMCI+ for lagged relations

Ordinary PC treats rows as independent, but in a time series the past of a variable is related to its present. Relations of interest are

$$X^j_{t-\tau} \longrightarrow X^k_t, \qquad \tau = 0, 1, \ldots, \tau_{\max}.$$

Algorithm: PCMCI+ (outline)

1. Among the past values of all variables, first reduce the candidate parents that may be related to each present value (a PC-type condition-selection step on lagged links).
2. Re-test each candidate link conditioning on the other past parents of both endpoints (momentary conditional independence, MCI), which removes autocorrelation effects.
3. Distinguish time-lagged relations from same-time (contemporaneous) relations and represent them in a time-series graph; contemporaneous links are oriented with PC-type rules.

Example output: $X^1_{t-2} \to X^2_t$ and $X^2_{t-1} \to X^3_t$. Feeding time series into ordinary PC can create false edges through autocorrelation: two independent but strongly autocorrelated series often show large sample correlations. Check the maximal lag $\tau_{\max}$ and stationarity.

3.3 Feedback: the DAG assumption fails

Economic and biological systems can have feedback such as $X_1 \to X_2 \to X_3 \to X_1$.

Method	Core idea
CCD	uses conditional independences to find relations common to the equivalence class of cyclic graphs
cyclic LiNGAM	uses linearity and non-Gaussian errors to further identify the cyclic structure

A linear cyclic SEM $X = BX + \epsilon$ is read as an equilibrium of a dynamic system, requiring $I - B$ invertible (stability typically needs the spectral radius of B below 1). CCD allows cycles, but directions the observational data cannot distinguish remain undetermined. Before using a cyclic algorithm, decide whether the feedback is *simultaneous* or actually a *time-lagged* relation that PCMCI+ should handle.

4 Choosing a Method for the Data Structure

Feature of the data	Method	Main points to check
hidden common causes suspected	FCI or RFCI	read PAG circles and arrowheads separately
nonlinear relations expected	PC/FCI with nonparametric tests	test power and computing time
time lags and autocorrelation	PCMCI+	maximal lag and stationarity
simultaneous feedback possible	CCD or cyclic LiNGAM	distinguish cycles from time lags

Summary

- With hidden confounders, PC's edges are not direct relations; FCI outputs a PAG whose arrowheads mean "not an ancestor" and whose circles mean "undetermined".
- FCI widens the search for separating sets (Possible-D-Sep) because separating sets may lie outside the neighbours; RFCI trades some information for speed.
- Nonlinear relations need kernel or distance-based CI tests; time series need lag-aware methods (PCMCI+); feedback needs cyclic methods (CCD, cyclic LiNGAM).

Chapter 12. Identifiable Causal Discovery Models: LiNGAM

Graphical models, Lecture 5 (non-Gaussian errors, residual independence, ICA-LiNGAM, DirectLiNGAM, variants)

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1 Directions That Conditional Independence Cannot Distinguish

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

All three structures satisfy $X \perp\!\!\!\perp Z \mid Y$. Using conditional independence alone we find the skeleton $X - Y - Z$ but cannot tell the three directions apart (Ch. 9). Structural equations can add information:

$$X = BX + \epsilon, \qquad \epsilon_j \text{ independent.}$$

The *distribution* of the errors, not only their covariance, can carry direction information.

2 The LiNGAM Model

Definition: LiNGAM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad X = BX + \epsilon.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering of variables, B is strictly lower triangular
independent errors	$\epsilon_1, \ldots, \epsilon_p$ are mutually independent: no hidden common cause among the analysed variables
non-Gaussianity	the structural errors are not normally distributed, which leaves information that distinguishes directions

Theorem: identifiability of LiNGAM (Shimizu et al., 2006)

Under linearity, acyclicity, mutually independent errors and non-Gaussian errors (at most one Gaussian error is allowed), the coefficient matrix B — hence the DAG and the causal order — is uniquely determined by the distribution of X.

3 In the True Direction, Cause and Regression Residual Are Independent

3.1 Two variables

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X \perp\!\!\!\perp \epsilon_Y, \quad \epsilon_X, \epsilon_Y \text{ non-Gaussian.}$$

Regressing Y on X leaves the residual $r_Y = Y - bX = \epsilon_Y$, so $X \perp\!\!\!\perp r_Y$.

Reverse direction $Y \to X$: fit $X = cY + r_X$. Then Y and r_X generally both contain ϵ_X and ϵ_Y, so for non-Gaussian data typically $Y \not\perp\!\!\!\perp r_X$.

Theorem: Darmois–Skitovich

Let $\epsilon_1, \dots, \epsilon_m$ be independent and $U = \sum_i \alpha_i \epsilon_i$, $V = \sum_i \beta_i \epsilon_i$. If $U \perp\!\!\!\perp V$, then every ϵ_i with $\alpha_i \beta_i \neq 0$ is Gaussian.

Theorem: the reverse regression leaves a dependent residual

In the model above with $b \neq 0$ and $\sigma_X^2 = \mathrm{Var}\epsilon_X$, $\sigma_Y^2 = \mathrm{Var}\epsilon_Y$, the least-squares coefficient is $c = \mathrm{Cov}(X, Y)/\mathrm{Var}Y = b\sigma_X^2/(b^2\sigma_X^2 + \sigma_Y^2)$ and

$$r_X = X - cY = \frac{\sigma_Y^2}{b^2\sigma_X^2 + \sigma_Y^2}\,\epsilon_X - c\,\epsilon_Y, \qquad Y = b\,\epsilon_X + \epsilon_Y.$$

r_X is uncorrelated with Y but, if either error is non-Gaussian, not independent of it.

Proof. $r_X = (1-cb)\epsilon_X - c\epsilon_Y$ and $1 - cb = \sigma_Y^2/(b^2\sigma_X^2 + \sigma_Y^2)$. Both coefficients of r_X are nonzero, and both coefficients of Y are nonzero. If $r_X \perp\!\!\!\perp Y$, Darmois–Skitovich forces ϵ_X and ϵ_Y to be Gaussian. Uncorrelatedness is the normal equation of least squares. ■

If instead both errors are Gaussian, (r_X, Y) is jointly Gaussian and uncorrelated, hence independent: *both* directions produce independent residuals, which is exactly why Gaussian SEMs are not identifiable. **LiNGAM regresses in both directions and picks the direction in which explanatory variable and residual are more independent.**

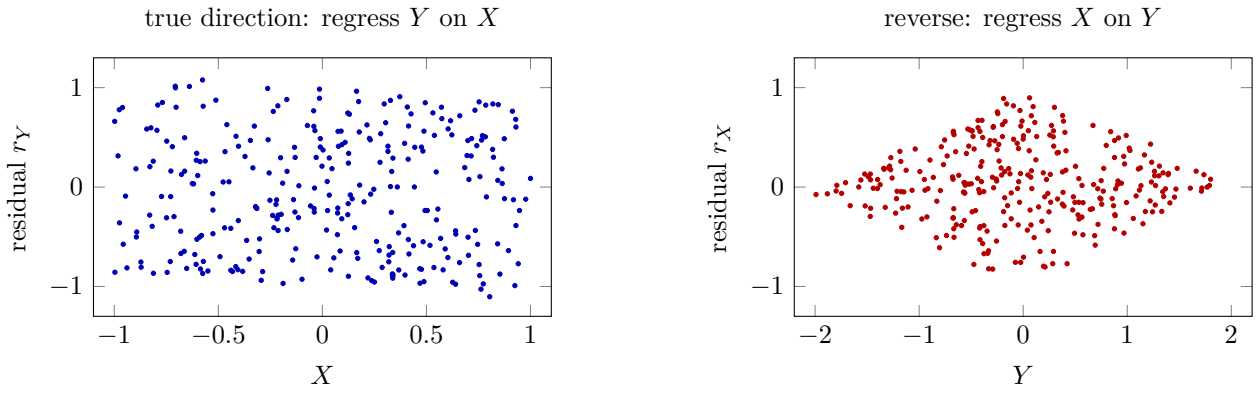


Figure 1: Simulation with $X = \epsilon_X$, $Y = X + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \mathrm{Unif}(-1, 1)$, $n = 300$. Both residuals are uncorrelated with their regressor (least squares guarantees it). In the true direction the cloud is a rectangle: $r_Y = \epsilon_Y$ is independent of X. In the reverse direction the cloud is a diamond whose vertical spread shrinks at the extremes of Y: uncorrelated but dependent (the correlation of the squares is 0.07 versus −0.42).

Example: a simulation picture

Take $\epsilon_X, \epsilon_Y \sim \mathrm{Unif}(-1, 1)$ and $Y = X + \epsilon_Y$. The scatterplot of (X, r_Y) is a rectangle (independent), while (Y, r_X) forms a rotated square (diamond) whose vertical spread of r_X shrinks at the extremes of Y: uncorrelated but clearly dependent.

4 Estimation

4.1 ICA-LiNGAM

Since B is permutable to strictly lower triangular form, $X = (I - B)^{-1}\epsilon = A\epsilon$: X is a linear mixture of independent non-Gaussian sources, the setting of independent component analysis (ICA).

Algorithm: ICA-LiNGAM

1. Run ICA to estimate an unmixing matrix W with WX having independent components; W equals $I - B$ up to an unknown row permutation and row scaling.
2. Find the row permutation of W giving no zeros (in practice, no small entries) on the diagonal; because $I - B$ has a unit diagonal and is triangular after ordering, this permutation is unique.
3. Divide each row by its diagonal element to get W', and set $\hat{B} = I - W'$.
4. Find the causal order by permuting $\hat{B}$ to be as close as possible to strictly lower triangular; prune small coefficients (e.g. by bootstrap or adaptive Lasso).

ICA is an iterative non-convex optimisation and can converge to local optima; the result can depend on initial values.

4.2 DirectLiNGAM finds exogenous variables one by one

Theorem: characterisation of an exogenous variable

In a LiNGAM, let $r_i^{(j)} = X_i - \dfrac{\mathrm{Cov}(X_i, X_j)}{\mathrm{Var}(X_j)} X_j$ for $i \neq j$. Then X_j is exogenous (has no parents) iff $X_j \perp\!\!\!\perp r_i^{(j)}$ for all $i \neq j$.

Proof ($\Rightarrow$) and idea ($\Leftarrow$). If X_j is exogenous, $X_j = \epsilon_j$. Every $X_i = a_{ij}\epsilon_j + \sum_{l \neq j} a_{il}\epsilon_l$ with $A = (I - B)^{-1}$, and $\mathrm{Cov}(X_i, X_j)/\mathrm{Var}(X_j) = a_{ij}$, so $r_i^{(j)} = \sum_{l \neq j} a_{il}\epsilon_l$, independent of ϵ_j. Conversely, if X_j has a parent, then X_j contains the error of some ancestor, and some residual $r_i^{(j)}$ shares a non-Gaussian error with nonzero coefficients with X_j; Darmois–Skitovich then gives dependence (Shimizu et al., 2011). ■

Moreover, after regressing out an exogenous variable, the residuals $(r_i^{(j)})_{i \neq j}$ again follow a LiNGAM with the remaining causal order, so the procedure can be repeated.

Algorithm: DirectLiNGAM

For each candidate X_j, regress every other X_i on X_j to obtain residuals $r_i^{(j)}$, and add up the dependence between the candidate and all residuals:

$$T(j) = \sum_{i \neq j} \mathrm{Dep}(X_j, r_i^{(j)}).$$

1. Evaluate every variable as an exogenous candidate.
2. Choose the variable with the smallest $T(j)$ as the first in the order.
3. Replace the remaining variables by the residuals with the chosen variable's effect removed.
4. Repeat the same process on the remaining variables.

Dep is a measure of statistical dependence, e.g. a kernel-based independence measure or a mutual-information estimate. Unlike ICA, DirectLiNGAM has no iterative non-convex step and, given a consistent dependence measure, converges to the correct order in a fixed number of steps $(p - 1)$.

4.3 Repeating residual independence yields the causal order

Example: three variables

Let the data come from the LiNGAM

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3,$$

with $\epsilon_1, \epsilon_2, \epsilon_3$ mutually independent non-Gaussian errors.

Step	Independence observed	Choice
1	X_1 is independent of the residuals of X_2, X_3 after removing X_1	choose X_1 first
2	after removing X_1's effect, X_2 is independent of the residual of X_3	choose X_2
3	the last remaining variable	choose X_3

Result: $X_1 < X_2 < X_3$. (For candidate X_2 in step 1, the residual of X_1 on X_2 shares ϵ_1, ϵ_2 with X_2, so $T(2) > T(1)$.)

4.4 After the order: estimate direct edges and effects

Suppose DirectLiNGAM found $X_1 < X_2 < X_3$. Each variable uses only the variables before it as candidate parents:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

In the example $\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{32} \approx 0.7$, $\hat{\beta}_{31} \approx 0$, so the estimated graph is

$$X_1 \Rightarrow X_2 \Rightarrow X_3.$$

Coefficients are estimated by least squares within the order, and small ones are pruned by sparse regression (adaptive Lasso) or tests; the order guarantees acyclicity.

5 Variants for Different Data Structures

Data or problem	Method	Caveat
independent cross-sectional data	DirectLiNGAM, ICA-LiNGAM	linear, acyclic structure, no hidden confounding
repeated measurements over time	VAR-LiNGAM	distinguish lagged and contemporaneous relations
feedback cycles	cyclic LiNGAM family	the acyclicity assumption of basic LiNGAM cannot be used
hidden common causes possible	ParceLiNGAM, RCD	estimate the identifiable part of the relations rather than the whole DAG

The LiNGAM variants are not applied one after another to patch a basic result; they are **chosen to match the data-generating structure**.

Remark: when LiNGAM is fragile

Near-Gaussian errors give weak signals, so large samples are needed. Measurement error, nonlinear effects, hidden confounders or selection bias break the independence of residuals and can make the wrong direction look better. Check the robustness of the order with

bootstrap resampling, and test the residual independence of the final model.

Summary

- Conditional independence leaves directions inside the Markov equivalence class undecided; non-Gaussian independent errors break the tie.
- In the true direction the cause is independent of the regression residual; in the reverse direction residual and regressor are uncorrelated but dependent (Darmois–Skitovich). With Gaussian errors both directions look the same.
- ICA-LiNGAM recovers B from the ICA unmixing matrix; DirectLiNGAM finds exogenous variables one by one through residual independence, then estimates coefficients within the order.

Chapter 13. Identifiable Gaussian Linear SEMs

Graphical models, Lecture 6 (equal error variances, conditional-variance ordering, ANM, the US algorithm)

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1 In General Gaussian SEMs, Directions Are Not Identified

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

The three structures can produce the same conditional independences and the same Gaussian distribution (Ch. 9).

General Gaussian SEM: $\epsilon_j \sim N(0, \sigma_j^2)$	Equal-variance Gaussian SEM: $\epsilon_j \sim N(0, \sigma^2)$
if each variable may have its own error variance, several directions are compatible with the same distribution result: CPDAG	if all structural errors are restricted to the same variance, the residual-variance structure differs by direction result: DAG

LiNGAM broke the tie with non-Gaussianity (Ch. 12). This chapter shows that, even for Gaussian data, an assumption on the *error variances* can identify the full DAG.

2 The Equal-Variance Gaussian SEM

Definition: equal-variance Gaussian SEM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad \epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

$$X = BX + \epsilon, \qquad \Sigma = \sigma^2 (I - B)^{-1} (I - B)^{-\top}.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering, B is strictly lower triangular
independent errors	no hidden common cause among the analysed variables
common error variance	all structural errors have the same σ^2
nonzero edges	coefficients of the edges in the graph are not 0

Under these assumptions B and the DAG are identified from the observational distribution.

3 Two Variables: The Equal-Variance Assumption Picks the Direction

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

so the two structural errors have $\mathrm{Var}(\epsilon_X) = \mathrm{Var}(\epsilon_Y) = \sigma^2$.

Reverse direction $Y \to X$: rewrite the same distribution as $Y = \epsilon'_Y$, $X = cY + \eta$. Then

$$\mathrm{Var}(\epsilon'_Y) = (1 + b^2)\sigma^2, \qquad \mathrm{Var}(\eta) = \frac{\sigma^2}{1 + b^2}.$$

Proof. $\mathrm{Var}Y = b^2\sigma^2 + \sigma^2$. The reverse regression has $c = \mathrm{Cov}(X,Y)/\mathrm{Var}Y = b/(1+b^2)$ and residual variance $\mathrm{Var}(X \mid Y) = \sigma^2 - \dfrac{b^2\sigma^4}{(1+b^2)\sigma^2} = \dfrac{\sigma^2}{1+b^2}$. ■

The two reverse error variances are equal only if $(1+b^2)^2 = 1$, i.e. $b = 0$. **Both directions explain the Gaussian distribution, but only the true direction satisfies the common error variance.**

4 Conditional Variances Reveal the Causal Order

4.1 The key lemma

Theorem: conditional-variance ordering

In an equal-variance Gaussian SEM, let S be an *ancestral* set (it contains all ancestors of each of its members, e.g. the first few variables of a causal order) and $j \notin S$. Then

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \quad \text{if } \mathrm{pa}(j) \subseteq S, \qquad \mathrm{Var}(X_j \mid X_S) \geq \sigma^2(1 + b_{jk}^2) > \sigma^2 \quad \text{otherwise,}$$

for a suitable parent $k \notin S$ of j.

Proof. Write $X = A\epsilon$ with $A = (I - B)^{-1}$; a_{jl} is the total effect of l on j (sum over directed paths of products of coefficients), $a_{jj} = 1$, and $a_{jl} = 0$ unless l is an ancestor of j or $l = j$. Because S is ancestral, every X_s, $s \in S$, is a function of $(\epsilon_l)_{l\in S}$ only, and conversely $(\epsilon_l)_{l\in S}$ is recovered from X_S (invert the triangular system). Hence, with $X_j = \sum_{l\in S} a_{jl}\epsilon_l + \sum_{l\notin S} a_{jl}\epsilon_l$, the second sum is independent of X_S and the first is determined by it:

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \sum_{l\notin S} a_{jl}^2 = \sigma^2\Big(1 + \sum_{l\notin S,\, l\neq j} a_{jl}^2\Big).$$

If $\mathrm{pa}(j) \subseteq S$, every ancestor of j is in S (ancestral), so the sum is empty. Otherwise let k be the parent of j outside S that is *latest* in a causal order. Any directed path from k to j other than the edge $k \to j$ ends with an edge $m \to j$ where m is a parent of j and a descendant of k; such an m is not in S (S is ancestral and $k \notin S$) and is later than k, contradicting the choice of k. So $a_{jk} = b_{jk} \neq 0$, giving the bound. ■

So, given the already ordered variables, a *source* of the remaining graph has the smallest possible conditional variance σ^2, and every non-source has strictly more. This proves:

Theorem: identifiability (Peters and Bühlmann, 2014)

The DAG and B of an equal-variance Gaussian SEM are identified from Σ. A causal order is obtained top-down: $\pi_1 = \arg\min_j \mathrm{Var}(X_j)$, and $\pi_m = \arg\min_{j\notin S} \mathrm{Var}(X_j \mid X_S)$ with $S = \{\pi_1, \ldots, \pi_{m-1}\}$.

4.2 A worked example

Example: three variables

Consider the Gaussian SEM

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3, \qquad \epsilon_1, \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0,1).$$

Step	Variance comparison	Choice
1	$\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = 1.64$, $\mathrm{Var}(X_3) = 1.8036$	X_1
2	$\mathrm{Var}(X_2 \mid X_1) = 1$, $\mathrm{Var}(X_3 \mid X_1) = 1.49$	X_2
3	the last remaining variable	X_3

Here $\mathrm{Var}(X_2) = 0.64 + 1$, $\mathrm{Var}(X_3) = 0.49 \times 1.64 + 1$, and $\mathrm{Var}(X_3 \mid X_1) = 0.49\,\mathrm{Var}(X_2 \mid X_1) + 1 = 1.49$. Result: $X_1 < X_2 < X_3$.

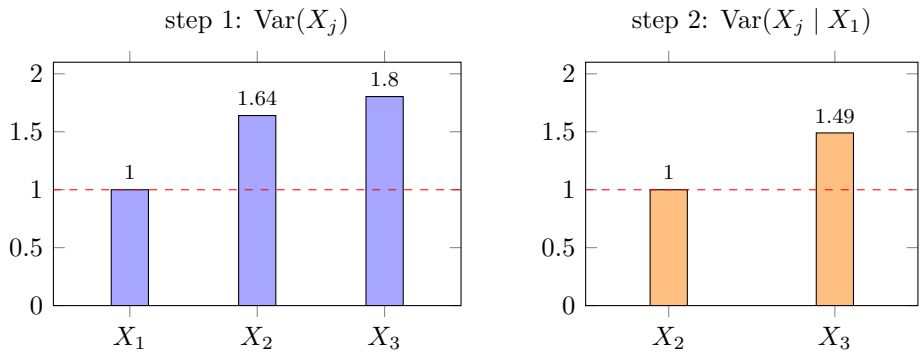


Figure 1: Top-down ordering in the equal-variance example ($\sigma^2 = 1$, red dashed line). At each step the variable whose (conditional) variance equals the common error variance — a source of the remaining graph — is selected: first X_1, then X_2.

4.3 After the order: select parents and edges

Let the estimated order be $X_1 < X_2 < X_3$. Each variable uses only earlier variables as candidate parents:

$$X_2 = \beta_{21} X_1 + \epsilon_2, \qquad X_3 = \beta_{31} X_1 + \beta_{32} X_2 + \epsilon_3.$$

Regression or sparse regression gives $\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{31} \approx 0$, $\hat{\beta}_{32} \approx 0.7$, hence

$$X_1 \to X_2 \to X_3.$$

Conditional variances find the order of the variables; regression coefficients select the direct edges within that order.

5 Equal Variance Does Not Mean Equal Observed Variances

Misconception	Correct meaning
all X_j must have the same variance	variation transmitted from the parents makes observed variances differ (as in the example)
standardising all variables gives equal variance	standardisation rescales the structural errors too, so it does not guarantee the equal-variance assumption — it typically destroys it
similar fitted residual variances prove the assumption	only a diagnostic compatible with the chosen DAG; it does not prove the true direction

When the equal-variance assumption is relatively natural: variables of the same kind and unit, whose measurement errors are also of similar size (e.g. expression levels measured on a common scale).

Remark: scale dependence and varsortability

The assumption is not scale-invariant: multiplying one variable by a constant changes its error variance. In many simulated benchmarks marginal variances simply increase along the causal order ("varsortability"), so methods exploiting variances can look excellent there; on real data with arbitrary units this signal may be absent or misleading. Check the variable units and measurement error, and the plausibility of the assumption, before trusting the order.

6 Additive Noise Models and the US Algorithm

6.1 The equal-variance Gaussian SEM is a special ANM

In an additive noise model each variable is

$$X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j, \qquad \epsilon_1, \dots, \epsilon_p \text{ independent.}$$

With S the set of already selected variables, define the expected conditional variance

$$U_j(S) = \mathbb{E}\{\mathrm{Var}(X_j \mid X_S)\}.$$

If S is ancestral and $j \notin S$, then $\epsilon_j \perp\!\!\!\perp X_S$ and

$$U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}].$$

Conditioning set S	Conditional variance
all parents are in S	$U_j(S) = \sigma_j^2$
some parent not yet in S	uncertainty created by the parents is added

6.2 ANMs are identified by an ordering of conditional variances

Let the true causal order be $\pi = (\pi_1, \dots, \pi_p)$ and $S_{m-1} = \{\pi_1, \dots, \pi_{m-1}\}$. If at each step the true next variable has a smaller conditional variance than the other candidates,

$$U_{\pi_m}(S_{m-1}) < U_k(S_{m-1}), \qquad k \notin S_{m-1} \cup \{\pi_m\},$$

then the variables can be identified one by one.

Equal-variance Gaussian SEM	General ANM
$\sigma_1^2 = \cdots = \sigma_p^2$	$\sigma_1^2, \dots, \sigma_p^2$ may differ
a variable whose parents are all included has no extra uncertainty, hence the smallest conditional variance (lemma above)	identified as long as the order of conditional variances, combining error variance and parent-induced uncertainty, is maintained

The identifiability of ANMs needs neither linearity nor Gaussianity, but the *correct conditional-variance order*. (For general ANMs with nonlinear f_j there are other identifiability results, based on residual independence as in LiNGAM; the conditional-variance route is the one used here.)

6.3 The US algorithm

Algorithm: US — order by conditional variances, then parents

Step 1: estimate the causal order. Let S be the selected set (initially $\varnothing$).

1. Regress each X_j, $j \notin S$, on X_S (flexibly, if f_j may be nonlinear).
2. Compute the residual variance $\widehat{U}_j(S)$.
3. Select the variable with the smallest value, $\hat{\pi}_m = \arg\min_{j \notin S} \widehat{U}_j(S)$.

Add the selected variable to S and repeat.
Step 2: select the parents. The candidate parents of each variable are the variables before it in the order. For each candidate, test conditional dependence given the other preceding variables:

$$X_{\pi_m} \not\perp\!\!\!\perp X_{\pi_j} \mid X_{\{\pi_1,\dots,\pi_{m-1}\}\setminus\{\pi_j\}}.$$

If conditional dependence remains, add the edge $\pi_j \to \pi_m$.

US finds the order of directions with conditional variances and selects the actual edges with conditional independence. Within a correct order, the preceding variables other than the parents are non-descendants, so the Markov property guarantees that non-parents are conditionally independent given the rest (and faithfulness that parents are not).

7 Summary

Model	Extra assumption	Clue for the direction
LiNGAM	linear relations with independent non-Gaussian errors	independence of the cause and the regression residual
equal-variance Gaussian SEM	common variance of independent Gaussian errors	order of conditional variances starting from the source
ANM and US	an additive noise model that keeps the conditional-variance order	conditional variances find the order; conditional independence selects the parents

ANM and US allow nonlinear functions and different error variances, but still need acyclicity, no hidden confounding and causal minimality.

Key points.

- A general Gaussian linear SEM cannot distinguish Markov-equivalent directions from observational data.
- If all structural errors have the same variance, the full DAG is identified from the observational distribution.
- The source has the smallest variance; repeatedly comparing conditional variances yields the causal order.
- After the order, regression or sparse regression estimates the parents and edge coefficients.
- Check variable scales, measurement error, hidden confounding and violations of the equal-variance assumption.
- The complete DAG is obtained not from Gaussianity alone but from the additional assumption of a common structural error variance.